

```

import java.util.*;

public class ContactManager {

    public static void main(String[] args) {

        HashMap<String, Contact> contacts = new HashMap<>();

        // Step 4: add contacts
        contacts.put("Ada Lovelace",    new Contact("Ada Lovelace",    "+1 617 555
0101"));
        contacts.put("Alan Turing",     new Contact("Alan Turing",     "+1 617 555
0102"));
        contacts.put("Grace Hopper",    new Contact("Grace Hopper",    "+1 617 555
0103"));
        contacts.put("Katherine Johnson", new Contact("Katherine Johnson", "+1 617
555 0104"));
        contacts.put("Linus Torvalds", new Contact("Linus Torvalds", "+1 617 555
0105"));

        // Step 5: look up a contact
        String lookupName = "Ada Lovelace";
        Contact found = contacts.get(lookupName);
        if (found == null) {
            System.out.println("Contact not found.");
        } else {
            System.out.println("Lookup result for \"" + lookupName + "\"");
            System.out.println(found);
        }

        // Test a name that doesn't exist
        String missingName = "Axle Rose";
        Contact missing = contacts.get(missingName);
        if (missing == null) {
            System.out.println("Lookup result for \"" + missingName + "\": Contact
not found.");
        } else {
            System.out.println(missing);
        }

        // Step 6: print sorted list
        ArrayList<Contact> sorted = new ArrayList<>(contacts.values());
        sorted.sort((a, b) -> a.getName().compareTo(b.getName()));

        System.out.println();
        System.out.println("*** All Contacts ***");
        for (Contact c : sorted) {
            System.out.println(c);
        }
    }
}

```

